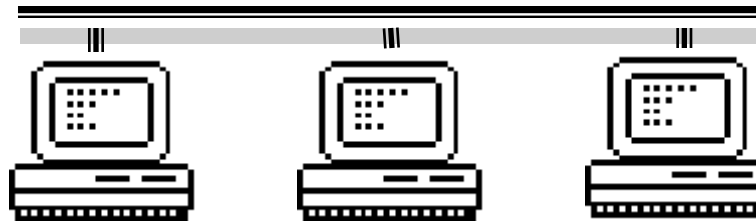


An overview of Database Management

Chapter 1



Today, Organizations Need...

Information to compete effectively

Information just to stay alive in the *information age*

Information organized in such a way that you can easily and quickly get to it

Information-processing tools that help you work with information

Applications of Database Systems

Supermarket:

bar code reader, price of the item, products database

Credit Card:

the purchases, credit limit, list of stolen or lost cards

Travel Agent:

holiday, flight and hotel details

University Library:

books, users and reservations

What you will learn

- 1 What is a database system ?
- 2 Why are database systems desirable?
- 3 Simple Operations on database
- 4 Components of the Database System
- 5 Types of data models
and the differences between them
- 6 Data Independence
- 7 Others

1 What is Database system?

(1)

A database system is basically just a computerized record-keeping system.

Electronic filing cabinet

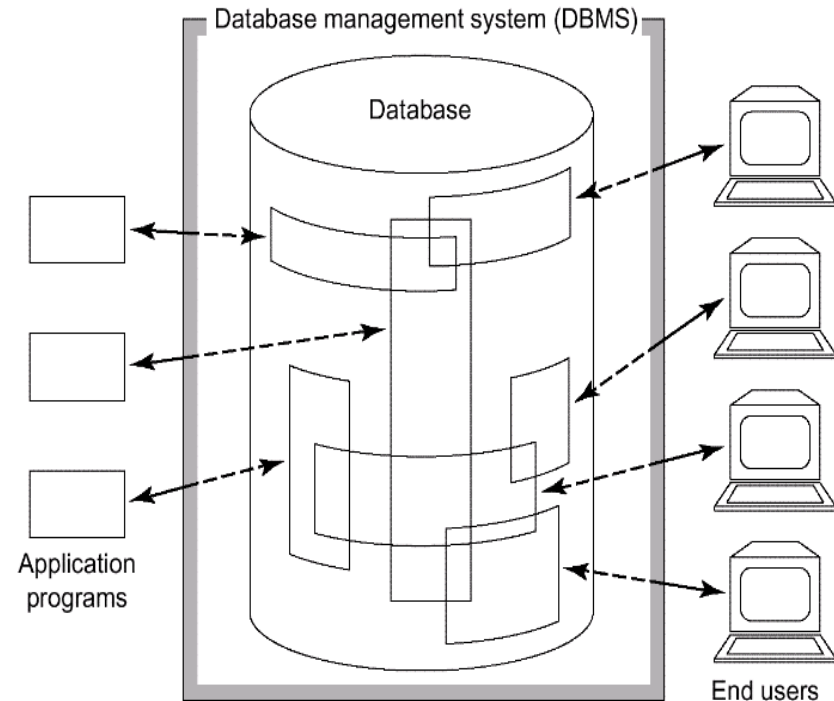


Fig. 1.4 Simplified picture of a database system

- computerized record-keeping system
- store information
- allow users to retrieve and update that information on demand

(2)

A **database system** is a shared collection of **logically related** data (and a description of this data), designed to meet the information needs of an organization.

Data and information ?

Logical deals with how knowledge workers view their information needs, and includes such terms as:

CHARACTER - our smallest unit of information.

FIELD - group of related characters.

RECORD - group of related fields.

FILE - group of related records.

DATABASE - group of logically associated files.

DATA WAREHOUSE - information from many databases.

An introduction to Database Systems

EMPLOYEE

EMPNO	FIRSTNME	MID INIT	LASTNAME	WORK DEPT	...
000010	CHRISTINE	I	HAAS	A00	
000020	MICHAEL	L	THOMPSON	B01	
000030	SALLY	A	KWAN	C01	
000050	JOHN	B	GEYER	E01	
000060	IRVING	F	STERN	D11	
000070	EVA	D	PULASKI	D21	

DEPARTMENT

DEPT NO	DEPTNAME	...
A00	SPIFFY COMPUTER SERVICE DIV.	
B01	PLANNING	
C01	INFORMATION CENTER	
D01	DEVELOPMENT CENTER	
D11	MANUFACTURING SYSTEMS	
D21	ADMINISTRATION SYSTEMS	



PROJECT

PROJNO	PROJNAME	DEPT NO	...
AD3100	ADMIN SERVICES	D01	
AD3110	GENERAL ADMIN SYSTEMS	D21	
AD3111	PAYROLL PROGRAMMING	D21	
AD3112	PERSONNEL PROGRAMMING	D21	
AD3113	ACCOUNT PROGRAMMING	D21	
IF1000	QUERY SERVICES	C01	

Relational Table: EMPLOYEE

EMPNO	FIRSTNME	MIDINIT	LASTNAME	...	BIRTHDATE	COMM
000010	CHRISTINE	I	HAAS	...	1933-08-24	9220
000020	MICHAEL	L	THOMSON	...	1948-02-02	0
000030	SALLY	A	KWAN	...	1941-05-11	-----
000050	JOHN	B	GEYER	...	1925-09-15	3214
.....				...		
000330	WING		LEE	...	1941-07-18	2030

row →

↑
column

↑
text column

↑
date column

↑
numeric column

Table : file

Rows: records

Columns : fields

Column data types: domain
: Primary key

(3)

A **database** is a collection of persistent data that is used by the application systems of some given enterprise

persistent data: be removed from the database only by some explicit request to the DBMS.

Enterprise examples:

- A manufacturing company
- A bank
- A hospital
- A university
- A government department
- A single individual with a small personal database

Student

Sno	Sname	Sage	Ssex	Sdept
020101	Chris	20	M	MIS
020102	Raman	19	M	MIS
020201	Lin Wang	20	F	CS
020202	Ming Zhang	19	M	CS

Data examples:

- Product data
- Customer data
- Patient data
- Student data
- Planning data
- Account data

(4)

A **database system** is really a collection of true propositions

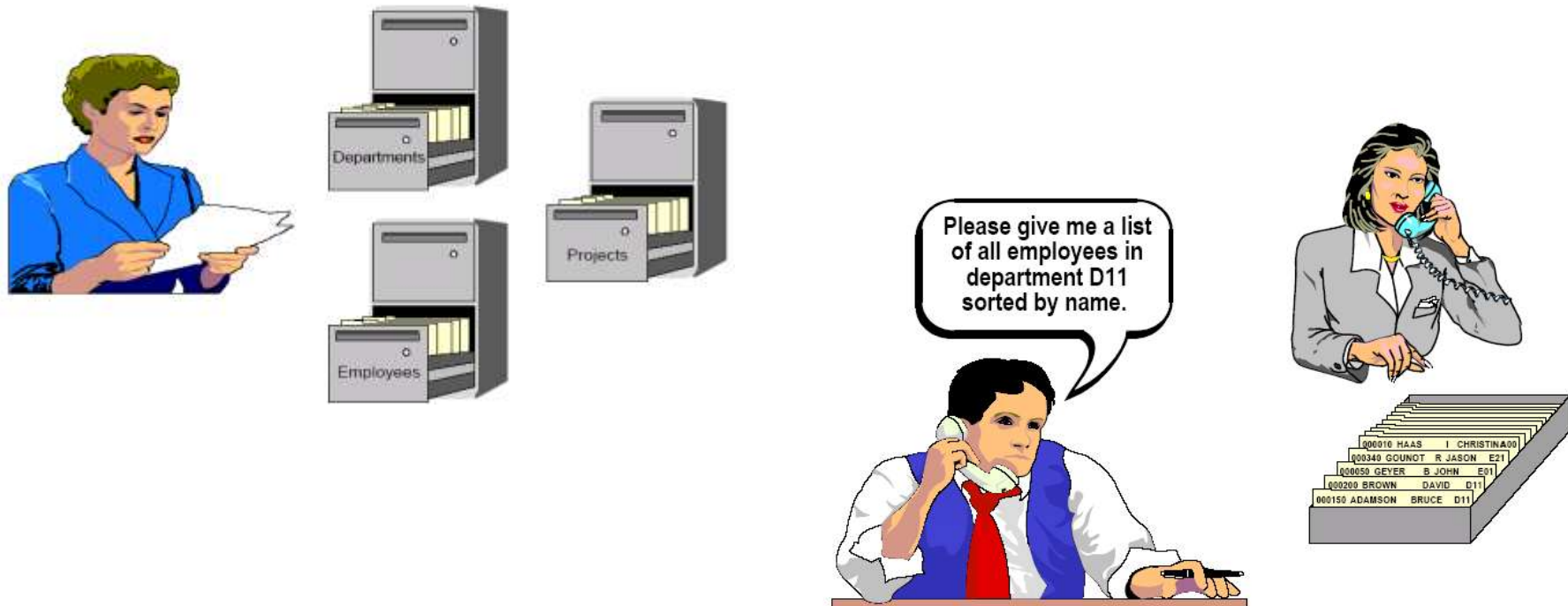
Data is given facts, from which additional fact can be inferred.

A given fact is called true proposition by logician.

2 Why database?

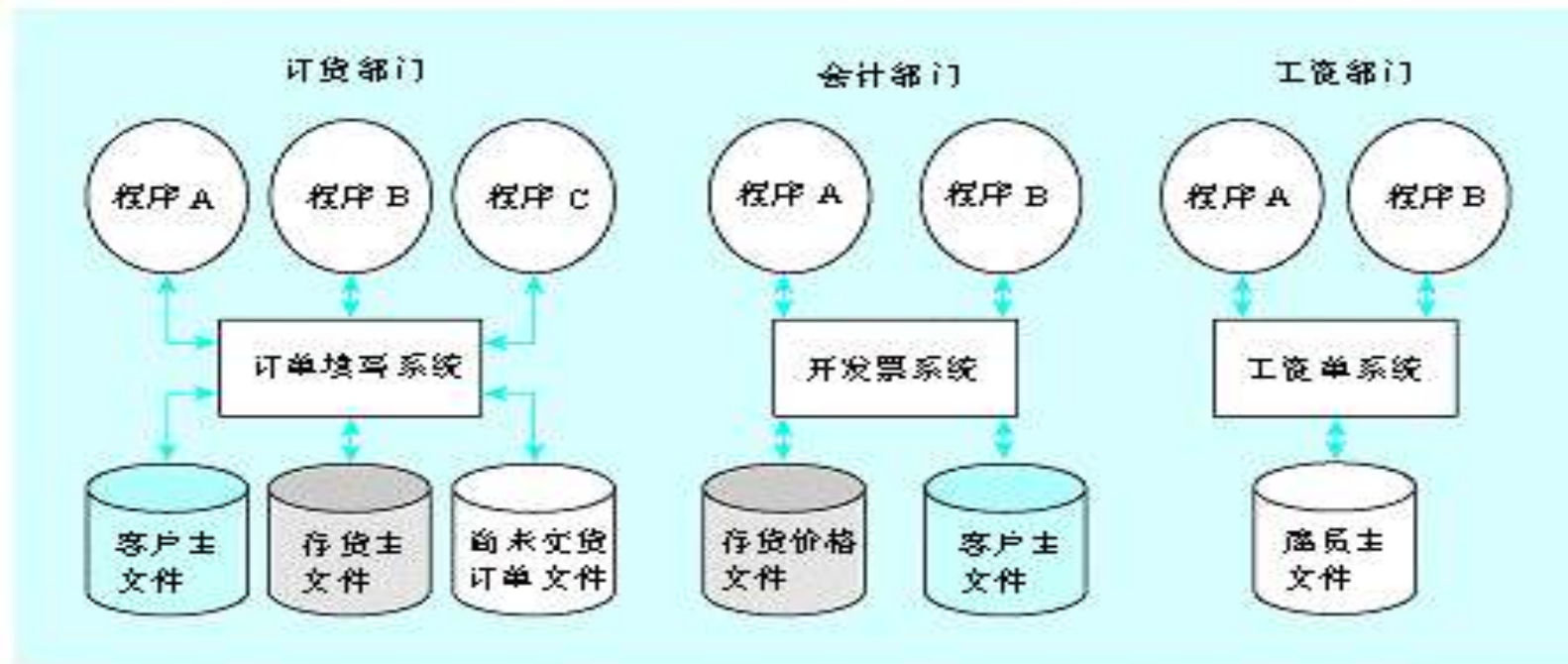
From point of view of development of data management technology

Computerless Data Storage



Traditional File-Based approach

A **file-based system** is a collection of application programs that performs services for the end-users such as the production of reports, but each program defines and manages its own data.



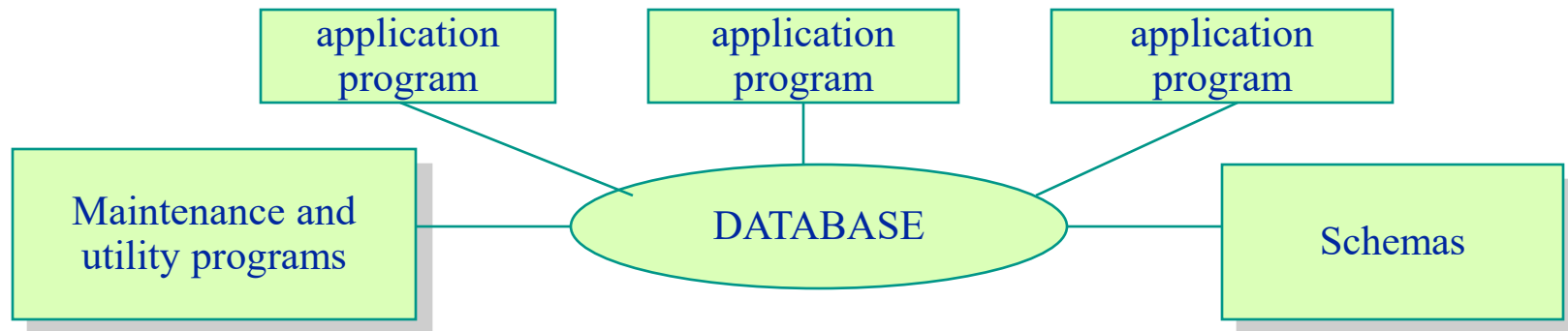
Limitations:

- ❑ Data are separated and isolated
- ❑ Data are often duplicated
- ❑ Application programs are dependent on file formats
 - Need to write a new program to carry out each new task
 - E.g. get the balance of accounts whose balance are less than 1000**
- ❑ Files are often incompatible with one another
- ❑ It is difficult to represent data in the user's perspectives
- ❑ Integrity problems
 - Integrity constraints (e.g. account balance > 0) become part of program code, Hard to add new constraints or change existing ones
- ❑ It is difficult to impose security on data
- ❑ Concurrent access by multiple users
 - Concurrent access needed for performance
 - Uncontrolled concurrent accesses can lead to inconsistencies
 - E.g. two people reading a balance and updating it at the same time**

 **Database systems offer solutions to all the above problems**

The Database Approach

The **database approach**: considering the data, instead of the application programs, of major importance in the design of systems -- i.e. look first at the design of the application's data and then write programs to process it!



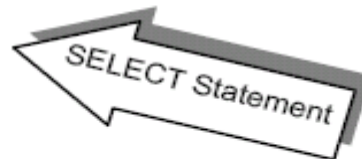
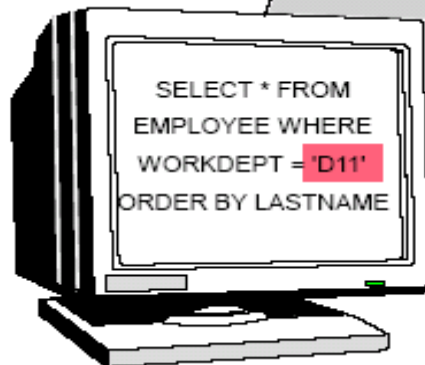
Schemas define the structure of records within the database
Maintenance and utility offer application-independent services, e.g. report generator.

Asking for Information - RDBMS

Return a list of
employees in
department D11
sorted by last name

RDBMS

EMPLOYEE					
EMPNO	LASTNAME	MID INIT	FIRSTNAME	WORKDEPT	PHONENO
000150	ADAMSON		BRUCE	D11	4510
000200	BROWN		DAVID	D11	4501
000050	GEYER	B	JOHN	E01	6789
000340	GOUNOT	R	JASON	E21	5698
000010	HAAS	T	CHRISTINE	A00	3978



Benefits of the database approach

The data can be shared

Redundancy can be reduced

Inconsistent can be avoided propagating updates

Transaction support can be provided transaction is a logical unit of work

Integrity can be maintained (ensuring the data in database is correct)

redundancy → inconsistency + other incorrect

Security can be enforced

Conflicting requirements can be balanced

Standards can be enforced

standardizing data representation is desirable as an aid to data interchange or movement between systems

data naming and documentation standards --- data sharing and understandability

You only see what you should

EMPNO	FIRSTNAME	MIDINIT	LASTNAME	WORKDEPT	PHONE
000100	BRUCE		ADAMSON	D11	4010
000200	DAVID		BROWN	D11	4001
000300	JOHN	B	GEYER	E01	6700
000400	JASON	R	GOUNOT	E21	5000
000500	CHRISTIE	T		ACC	3070

• Data security • Simplification

The advantages of a database system over file-based system

Compactness

Speed

Less drudgery

Current

Provides the enterprise with centralized control of
its data-----overriding advantage

3 Simple Operations on database (SQL)

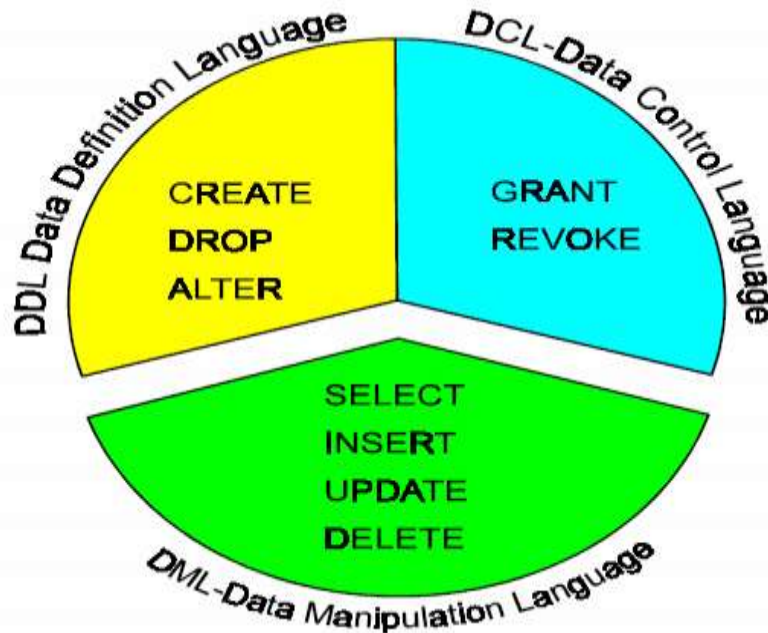
File---

adding removing

Record---

Insert Delete Select Update
(SQL)

Structured Query Language--SQL



1986年10月，美国**ANSI**采用**SQL**作为关系数据库管理系统的标准语言(**ANSI X3.135-1986**)，后为国际标准化组织(**ISO**)采纳为国际标准。

SQL has become one of the most important elements of modern database technology. It is used very extensively by virtually all database systems and is a major vehicle in facilitating inter-database communication

DDL

DDL: Specification notation for defining the database

E.g.

```
create table account (  
    account-number char(10),  
    balance integer)
```


DML

Language for accessing and manipulating the data

DML also known as query language

Two classes of languages

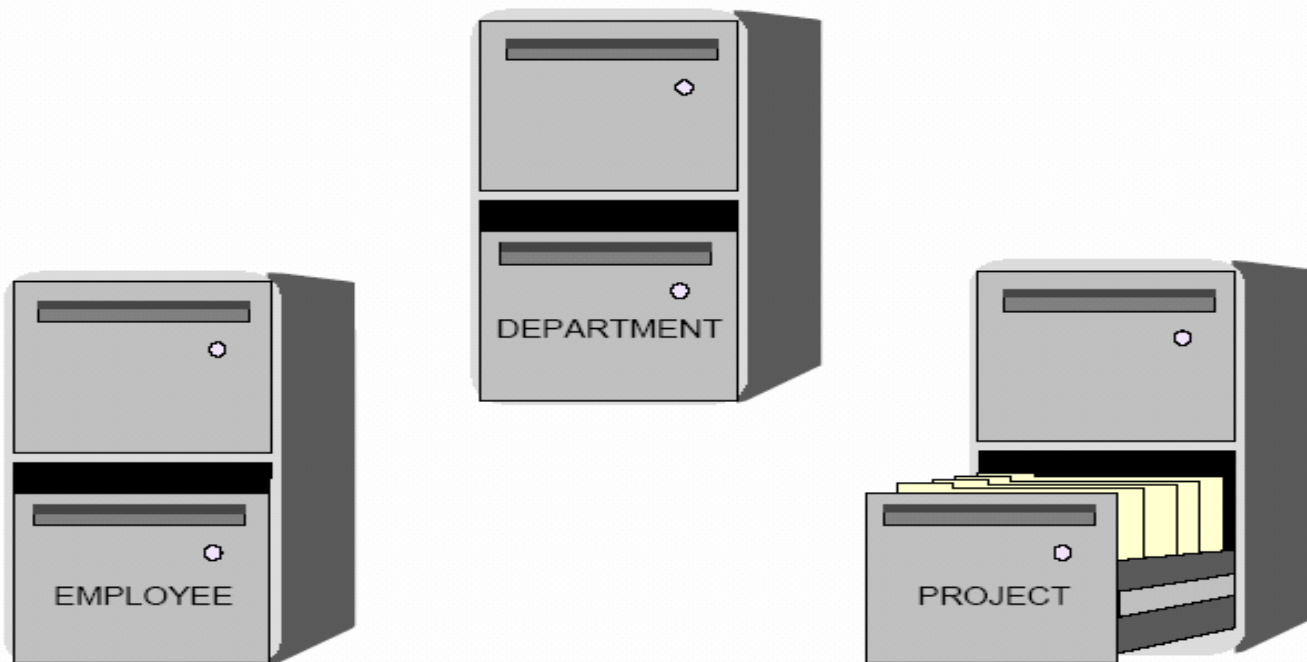
Procedural – user specifies what data is required and how to get those data

Nonprocedural – user specifies what data is required without specifying how to get those data

SQL is the most widely used non-procedural language

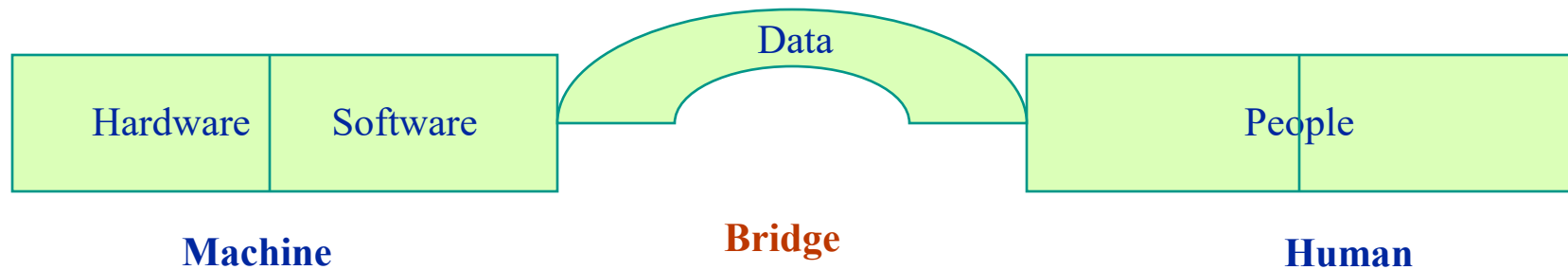
DCL

GRANT SELECT ON PROJECT TO SALLY

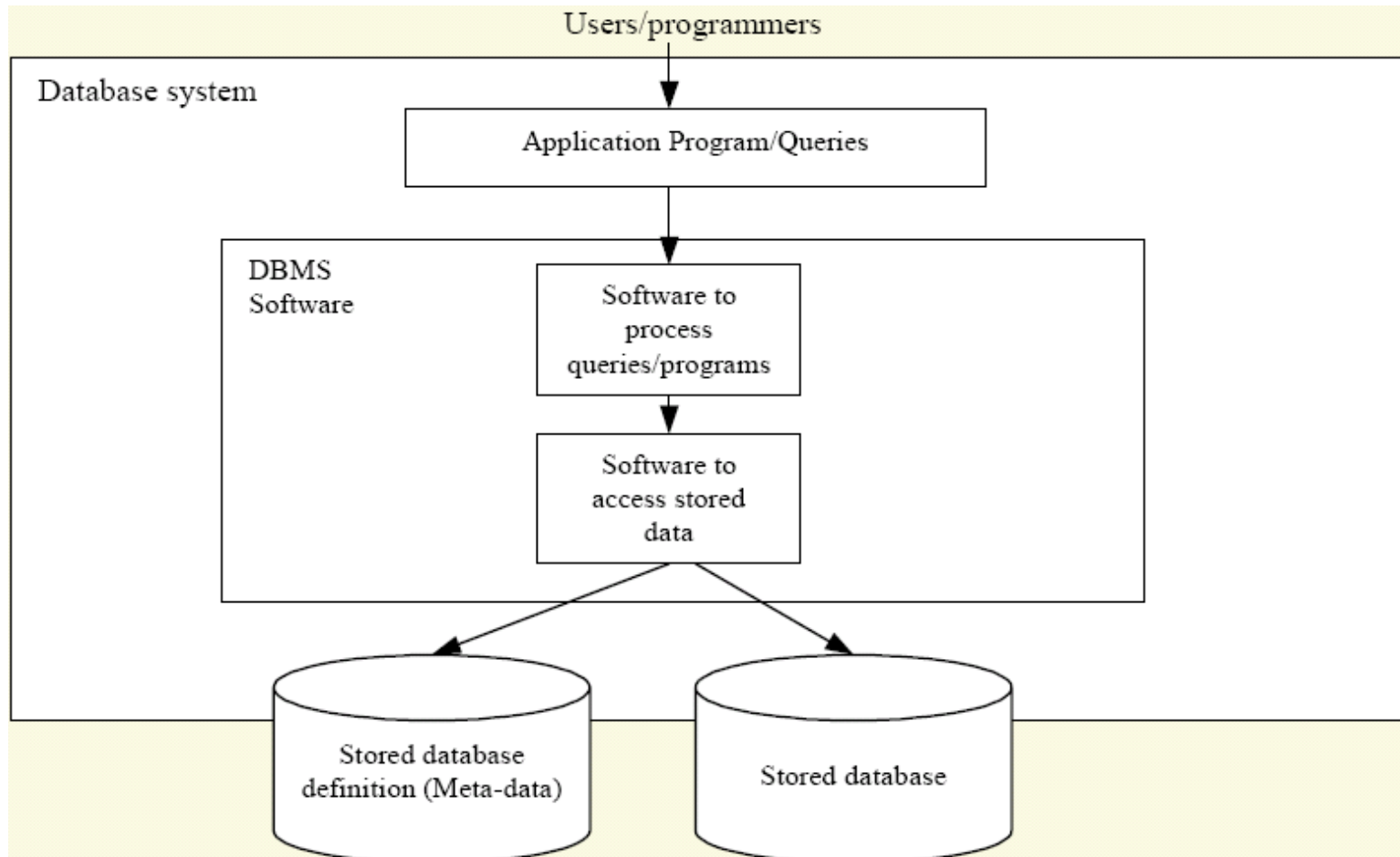


REVOKE SELECT ON PROJECT FROM SALLY

4 Components of the Database System



Database System



----Data

Data: refer to what is actually stored in the database
information refer to the *meaning* of that data as understood by some user.

The terms “data” and “information” are treated as synonymous(同义的) in this book.

EMPLOYEE	NAME	ADDRESS	DEPARTMENT	SALARY	...
ENROLLMENT	NAME	COURSE	...		

Fig. 1.5 The EMPLOYEE and ENROLLMENT files

Single-user system is a system in which at most one user can access the database at any given time.

Muti-user system is a system in which many users can access the database at the same time.

Two assumptions in our book: multi-user system, data store in a single DB

The data in database is

- ✓ Integrated.

(database can be thought of as a unification of distinct files with redundancy eliminated)

- ✓ shared $\Leftrightarrow$ personal or application-specific

- ✓ persistent $\Leftrightarrow$ ephemeral

----Hardware

Consisting of :

magnetic disks, disk drives: store information (the secondary storage)

Hardware processor(s) and associated main memory: running DBS software

----Software

Database Management System (DBMS).---the most important software

component

Eg., Sybase、 Oracle、 SQL Server、 Informix、 FoxBase、
FoxPro、 Access...

database package is used to refer to a DBMS implementation offered by a
particular vendor

Application development tools

Design aids

Report writer

Transaction manager or TP monitor

Database Management System (DBMS) is the software that manages and controls access to the database -- an integrated part of our day-to-day life that often we are not aware we are using one.

A DBMS contains 5 software components:

- DBMS engine

- Data definition subsystem

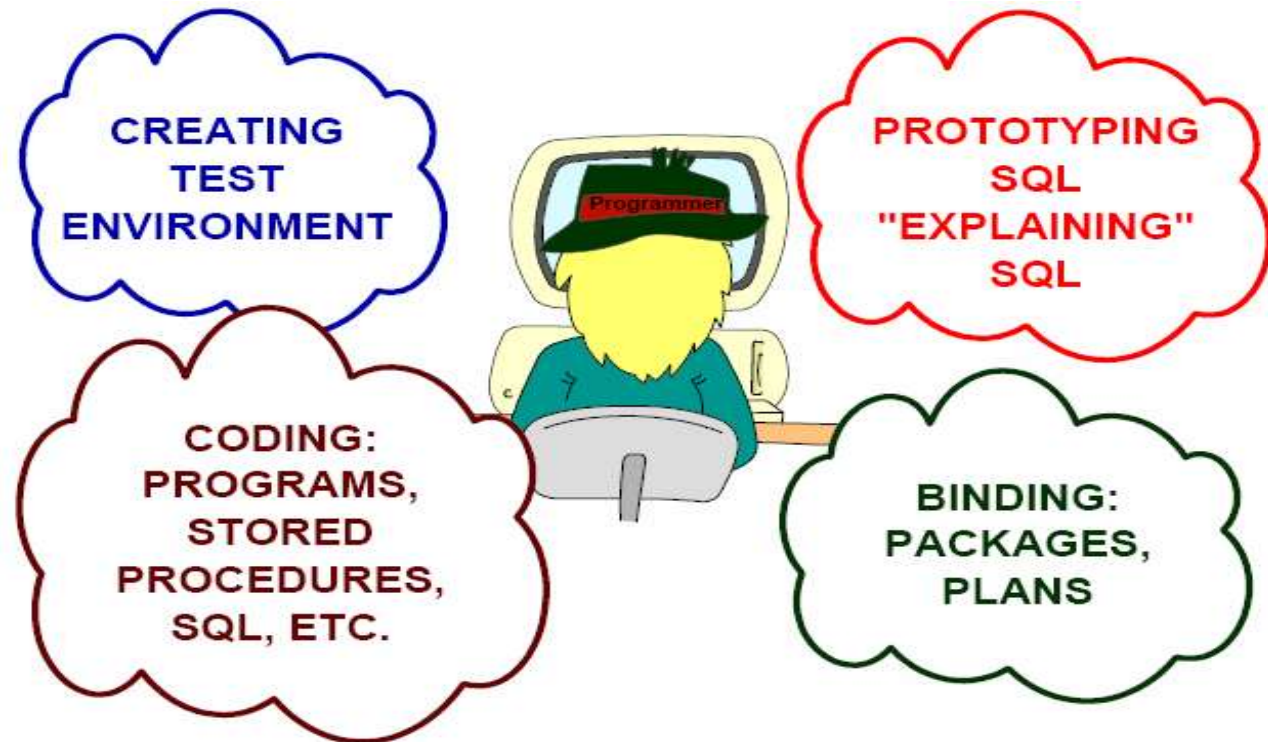
- Data manipulation subsystem

- Application generation subsystem

- Data administration subsystem

----Users

Application programmers
COBOL, PL/I, PB, JAVA...



■ End user

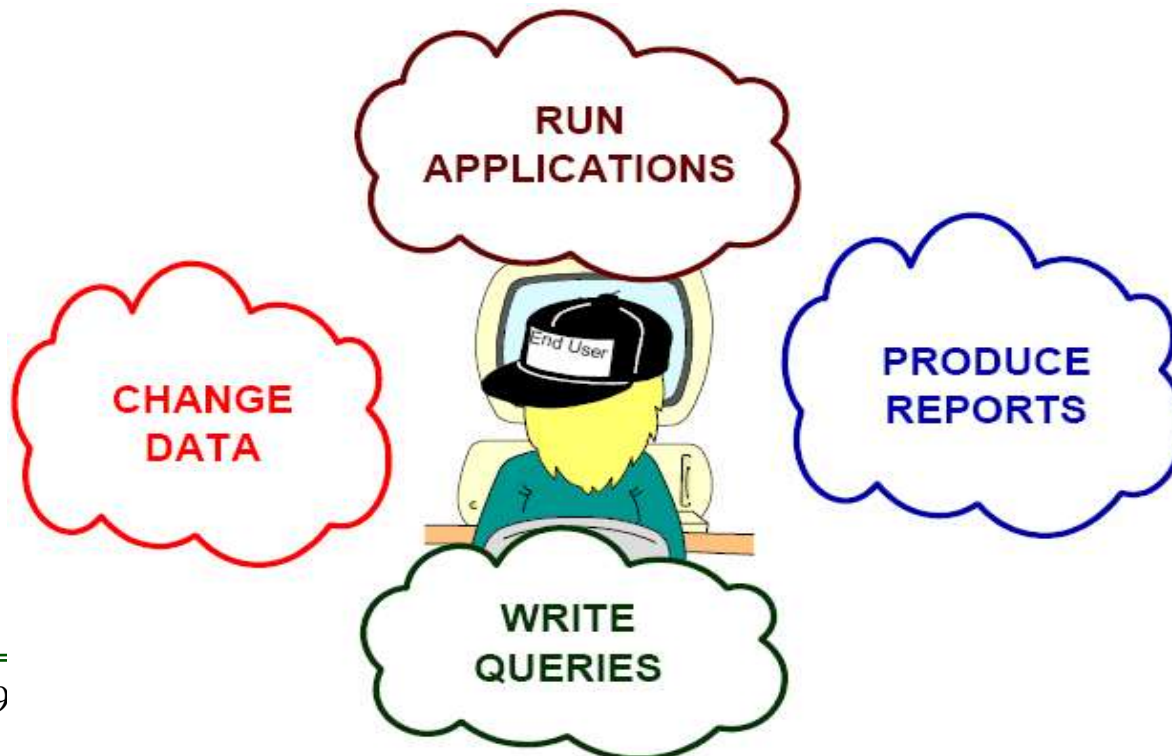
interacting with system from online workstations or terminals.

access data by: online applications

interface (Query language processor)

menu or forms-driven interfaces

command –driven interfaces



Database administrator(DBA)-- IT technician

1 responsible for **implementing** the data administrator's **decisions**.

2 **create the actual database** and to implement the **technical controls** needed to enforce the various policy decisions made by the data administrator.

DBA is responsible for administrating the DB and DB system in accordance with policies established by data administrator

Data Administrator (DA, 数据管理员) --manager

1 decide **what data** should be stored in the database in the first place

2 establish **policies** for maintaining and dealing with that data once it has been stored.

Data administrator -has central responsibility for data, understand the data and needs of enterprise

Database Administrator



5 Types of data models and the differences between them

Data model

A **data model** is an abstract, self-contained, logical definition of the objects, operators and so forth. That together constitute the abstract machine with which users interact

the object allow us to model data structure

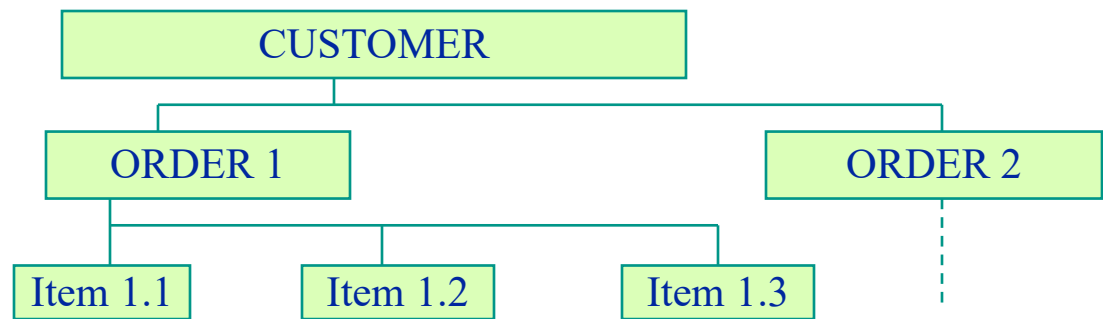
the operators allow us to model its behavior

Types of data models

database system are categorized according to data structures and operators they present to the user

1 Hierarchical Model

such as IBM's IMS
traversing points



Hierarchical Model Database

2 Network Model

Limitations:

Complex programs have to be written to answer even simple queries

There is minimal data independence

There is no widely accepted theoretical foundation

3 Relational model

data is represented as rows in tables(it is a true proposition)

Operators are provided for operating on rows in tables

is based on logic and mathematics (Why so dominant?)

4 object/relational model such as DB2 and informix

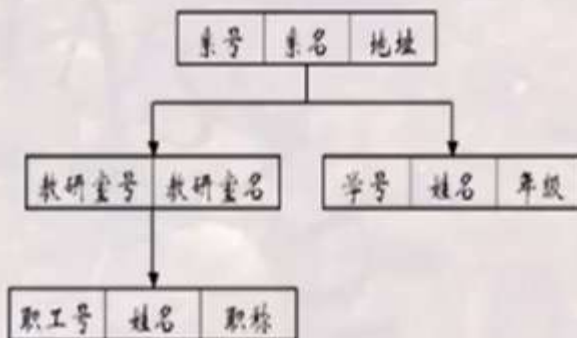
5 object model entirely different such as Gemstone and Versant ODBMS

Others are ad hoc, relational model is based on formal logic

层次模型的例子



层次模型
(树的形式)



层次模型的模式



层次模型的数据

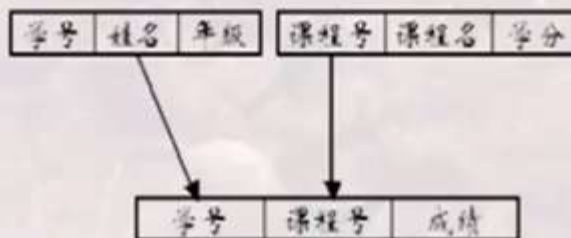
层次模型:

- 1.有且只有一个节点没有双亲节点，称为根节点;
- 2.根以外的其他节点有且只有一个双亲节点;
- 3.每个结点表示一个记录类型。

层次数据库系统只能处理一对多的实体联系。

层次结构采用关键字来访问其中每一层次的每一部分。

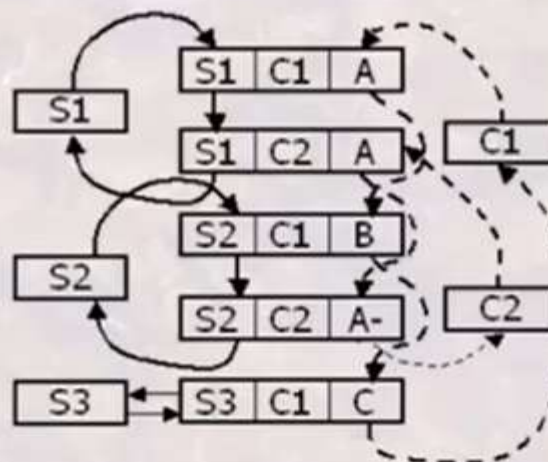
1973年, Charles W. Bachman: “网状数据库/DBTG之父”



网状模型的模式



网状模型
(图的形式)



网状模型的数据

网状模型:

1. 一个结点可以有多个双亲;
2. 结点直接可以有多种联系。

具有多对多类型的数据组织方式。

用连接指令或指针来确定数据间的显式连接关系。

应用发展要求实现数据模拟和行为模拟，传统数据库技术开始表现出明显不足：

- (1) 数据对象简单，只能检索一组记录和由同质记录共同组合的集合，无复杂的嵌套数据和复杂数据。
- (2) 对象之间的关系简单，不能实现实体间聚合、继承等复杂联系。
- (3) 一致约束不完全，只能预定时机检查。
- (4) 事务短寿，并发控制机制简单。

20世纪90年代来，在关系型数据库基础上，引入面向对象技术，从而使关系型数据库发展成为一种新型的面向对象关系型数据库。它始于工程和设计领域的应用，并且成为广受金融、电信和万维网应用欢迎的系统。它适用于多媒体应用以及复杂的、很难在关系DBMS里模拟和处理的关系。

面向对象的模型

采用面向对象的方法来设计数据库。面向对象的数据库存储对象是以对象为单位，每个对象包含对象的属性和方法，具有类和继承等特点。**Jasmine**就是面向对象模型的数据库系统。

- ✓ 面向对象数据模型的数据结构是非常容易变化的。对象模型没有单一固定的数据结构。编程人员可以给类或对象类型定义任何有用的结构，如链表、集合、数组等。此外，对象可以包含可变的复杂度，利用多重类型和多重结构。
- ✓ 对象模型可以用二维表来表示，称为对象表。但对象表是用一个类（对象类型）表定义的。一个对象表用来存储这个类的一组对象。对象表的每一行存储该类的一个对象（对象的一个实例），对象表的列则与对象的各个属性相对应。因此，在面向对象数据库中，表分为关系表和对象表，虽然都是二维表的结构，但却是基于两种不同的数据模型。

了解对象

对象是面向对象编程中重要概念，用对象来表示现实世界中的实体。一个学生、一门课程、一次考试记录都可以看作对象。每个对象包含一组属性和一组方法。

属性用来描述对象的状态、组成和特性，是对象的静态特征。一个简单对象如整数，其本身就是状态的完全描述，不需要其他属性，这样的对象称为原子对象。属性的值可以是复杂对象。一个复杂对象包含若干属性，而这些属性作为一种对象，又可能包含多个属性，这样就形成了对象的递归引用，从而组成各种复杂对象。

每一个对象都是其属性和方法的封装。用户只能见到对象封装界面上的信息，对象内部对用户是隐蔽的。封装的目的是为了使对象的使用和实现分开，使用者不必知道行为实现的细节，只需用消息来访问对象，这种数据与操作统一的建模方法有利于程序的模块化，增强了系统的可维护性和易修改性。

各模型优缺点

	层次模型	网状模型	关系模型	面向对象模型
优点	1、简单，命令少，易操作。2、实体间联系固定的应用系统，层次模型性能优于关系模型，不次于网状模型。3、良好完整性支持。	1、能更直接的描述现实世界。 2、具有良好性能，存取效率高。	1、建立在严格数学概念基础上。2、概念单一数据结构简单清晰。3、存取路径透明，数据独立性更高。	存储数据和操作。能定义嵌套结构，很强的模型扩展能力，节省开销。
缺点	表示非层次性不利。插入删除限制较多，查询子女结点须通过双亲结点，命令趋于程序化。	其DDL语言极其复杂，数据独立性差。	查询效率不如非关系模型，需要对查询优化。	技术尚不成熟，无完全支持的DBMS产品。

An implementation of a data model

---- is a physical realization on a real machine of the components of the abstract machine that together constitute that model

The model is what users have to know about
but the implementation is what users do not have to know about.

the distinguish of model and implementation is a special case of Logical and physical.

Gap between database principle (ought to be) and database practice(actually are)

6 Data Independence

Logical Data Independence

Physical Data Independence

Change the data physical representation and access technique without affecting the application.

Data physical representation: how the data is physically represented in storage

Eg1. EMPLOYEE file:

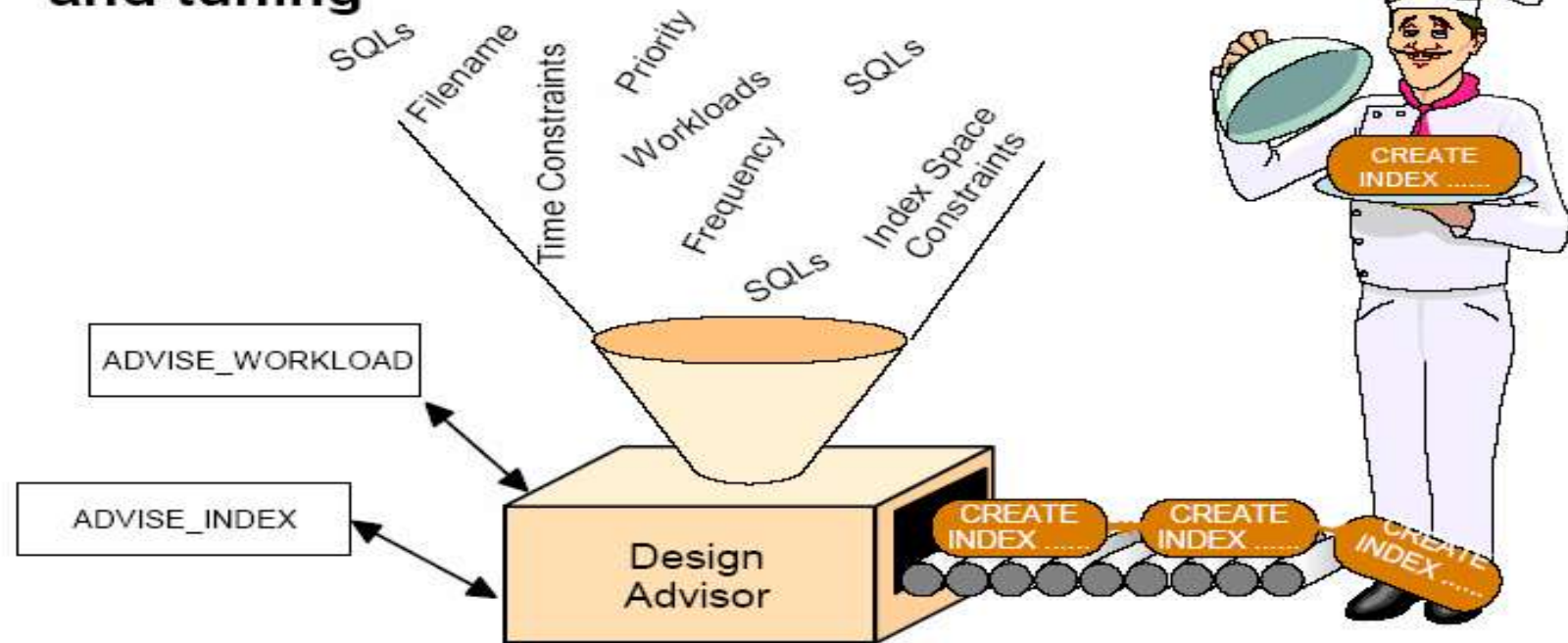
- **indexed** on its "employee name" field

Eg2. SALARY field of EMPLOYEE file: **decimal or binary**

Data access technique: how the data is physically accessed

Index

- **Assists in finding the right indexes**
 - Based on Workload
 - Virtual indexes
- **Reduce complexity of performance analysis and tuning**



Relation Scan

① SELECT *
FROM PHONEBOOK

② SELECT *
FROM PHONEBOOK
WHERE ADDRESS LIKE '%AVE%'

PHONEBOOK

LASTNAME	FIRSTNME	MIDINT	ADDRESS	PHONENO
Smith	Diana	A	2253 Dahlia St.	555-1111
Gaines	Lois	B	6094 Beaker St.	555-1221
Brown	Terry	S	45 River Walk	555-2168
Adams	Sandy	A	26 Cisco Lane	555-1311
Friedrich	John	P	5 Beaumont Rd.	555-1411
Jones	Larry	R	1921 Hill Rd.	555-3242
Wise	Sara	J	25 Rawlings St.	555-2345
Jones	Laverne	G	504 1st Ave	555-1777
Adams	Bobby	D	1426 Maple Dr.	555-1423
Jones	Larry	R	240 Boswell Dr.	555-5390
Smith	Grace	G	983 Famous Rd.	555-8764
Caldwell	Simone	B	122 42nd St.	555-5367

* Phonebook has an index on LASTNAME, FIRSTNME

Index-Only Access

```
SELECT LASTNAME, FIRSTNME, PHONENO  
FROM PHONEBOOK  
WHERE LASTNAME LIKE 'S%'
```

Root Page

Howell, Thurston (1)
Peters, Joan (2)
Zidler, Bob (3)

Leaf Pages

Adams, Mark (2,1)(5,2)
Alviani, Carl (7,2)
Bennett, Barb (2,3)
Chapman, Mike (5,1)
Davis, Mac (3,1)
Howell, Thurston (3,2)

Jefferson, Tom (4,1)
Jones, Bob (1,1)
Lee, Spike (4,2)
Lincoln, Abe (7,1)
Mitchell, David (2,3)
Peters, Joan (1,3)

Quinn, Tony (6,2)
Reeves, Susan (1,2)
Smith, Stanley (8,2)
Smith, Steve (8,1)
Thompson, Tom (6,3)
Zidler, Bob (6,1)

Data Pages

Jones, Bob
Reeves, Susan
Peters, Joan

Adams, Mark
Mitchell, David
Bennett, Barb

Davis, Mac
Howell, Thurston

Jefferson, Tom
Lee, Spike

Chapman, Mike
Adams, Mark

Zidler, Bob
Quinn, Tony
Thompson, Tom

Lincoln, Abe
Alvani, Carl

Smith, Steve
Smith, Stanley

*INDEX ON LASTNAME, FIRSTNME

Non-Clustered index

Root Page

Howell, Thurston (1)
Peters, Joan (2)
Zidler, Bob (3)

Leaf pages	Adams, Mark (2,1),(5,2)	Jefferson, Tom (4,1)	Quinn, Tony (6,2)
	Alviani, Carl (7,2)	Jones, Bob (1,1)	Reeves, Susan (1,2)
	Bennett, Barb (2,3)	Lee, Spike (8,1)	Smith, Stanley (7,1)
	Chapman, Mike (5,1)	Lincoln, Abe (8,2)	Smith, Steve (4,2)
	Davis, Mac (3,1)	Mitchell, David (2,2)	Thompson, Tommy (6,3)
	Howell, Thurston (3,2)	Peters, Joan (1,3)	Zidler, Bob (6,1)

Data pages	Jones, Bob	Adams, Mark	Davis, Mac	Jefferson, Tom
	Reeves, Susan	Mitchell, David	Howell, Thurston	Smith, Steve
	Peters, Joan	Bennett, Barb		
	Chapman, Mike	Zidler, Bob	Smith, Stanley	Lee, Spike
	Adams, Mark	Quinn, Tony	Alviani, Carl	Lincoln, Abe
		Thompson, Tommy		

Clustered index

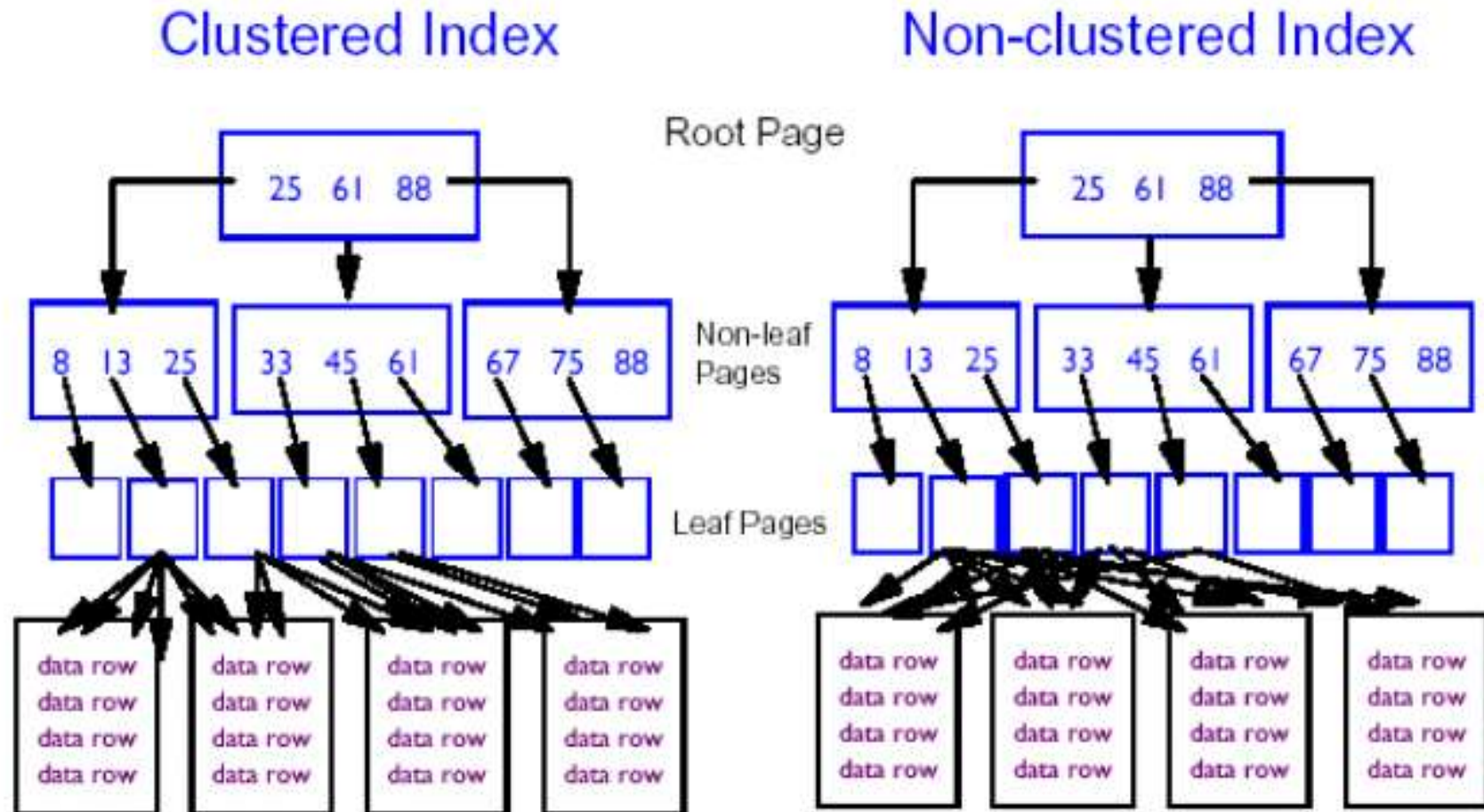
Root Page

Howell, Thurston (1)
Peters, Joan (2)
Zidler, Bob (3)

L e a g f e s	Adams, Mark (1,1),(1,2)	Jefferson, Tom (3,2)	Quinn, Tony (5,2)
	Alviani, Carl (1,3)	Jones, Bob (3,3)	Reeves, Susan (5,3)
	Bennett, Barb (2,1)	Lee, Spike (4,1)	Smith, Stanley (6,1)
	Chapman, Mike (2,2)	Lincoln, Abe (4,2)	Smith, Steve (6,2)
	Davis, Mac (2,3)	Mitchell, David (4,3)	Thompson, Tommy (6,3)
	Howell, Thurston (3,1)	Peters, Joan (5,1)	Zidler, Bob (7,1)

D P a g e s	Adams, Mark	Bennett, Barb	Howell, Thurston	Lee, Spike
	Adams, Mark	Chapman, Mike	Jefferson, Tom	Lincoln, Abe
	Alviani, Carl	Davis, Mac	Jones, Bob	Mitchell, David
	Peters, Joan	Smith, Stanley	Zidler, Bob	
	Quinn, Tony	Smith, Steve		
	Reeves, Susan	Thompson, Tommy		

Clustered and Non-Clustered Index Access



Data-independence is a major objective for Database system

For the following two reasons:

- Different applications will require different views of the same data
- The DBA must have the freedom to change the physical representation or access technique in response to changing requirements without having to modify existing applications.
eg: new kinds of data, new standards, application priorities, new storage devices.....year 2000

Data-independence can be defined as the immunity of applications to change in physical representation and access technique

Stored representation of data

Representation of numeric data:

decimal, character string,... (base,scale,mode,precision)

Representation of character data:

several distinct coded character sets—e.g., **ASCII**(American Standard Code for Information Interchange, **EBCDIC**(Extended Binary Coded Decimal Interchange Code), **Unicode** (统一码、万国码、单一码、标准万国码) .

Units for numeric data:

inches/centimeters

Data coding:

(part color="Red" or "Blue" or "Green" ...)

1 = "Red," 2 = "Blue," ...

Stored representation of data

Data materialization: constructing an occurrence of the logical field from the corresponding stored field occurrence and presenting it to the application

Direct:

single stored field instance



logical field



application

Indirect:

several stored field instance



computation

logical field

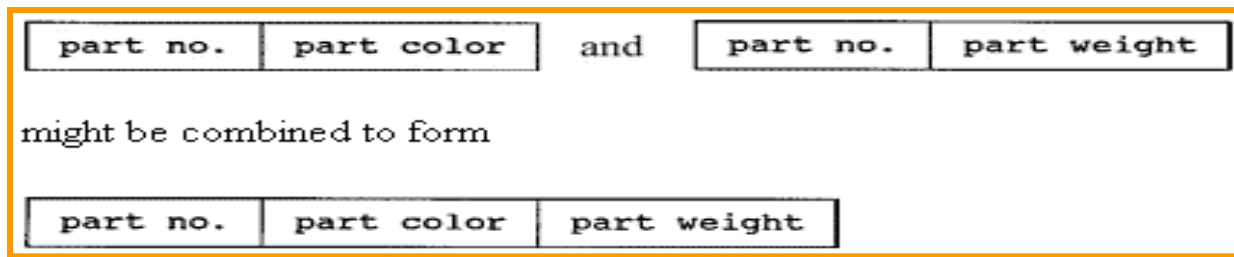


application

Stored representation of data

Structure of stored records:

Two existing stored records might be combined into one.



a single stored record type might be split into two.

Stored representation of data

Structure of stored files :

- A given stored file can be physically implemented in storage in a wide variety of ways.

a single disk

several different device

one or more indexes ,one or more embedded pointer chains (or both)

some hashing scheme

- ☞ DBA choose the stored representation of data.
- ☞ Won't affect applications, but performance

Physical data independence

Physical data independence: DBA make changes the stored representation of data while the data as seen by applications does not change.

- Basic way to provide data independence :
three-level architecture

7 Others.....

When not to use a database system?

DBMS requires **additional overhead!**

- ❑ High initial investment in hardware, software and training.
- ❑ Overhead for providing security, concurrency control, recovery and integrity functions.

☞ There are still a lot of applications that use files.

- The database and application are simple, well-defined, and **not expected to change** -- usually dealing with **small size of data** (better either in a file or main memory!)
- There are **stringent real-time requirements** for some programs that may not be met because of DBMS overhead.
- Multiple-user access to data is not required.

The major events in the database systems history

- 1961 The first generalized DBMS-GEs IDS by Bachman
- 1967: IBM 开发出ICS/DL/I, 信息控制系统与数据语言/界面, 这是阿波罗 (Apollo) 项目的分级数据库。ICS随后变成了IMS, 信息管理系统, 与IBM的System360主机整合到一起。
- 1970 The relational model is developed by Codd at IBM research Lab
- 1975 Entity-Relationship (ER) model introduced by Chen
- Research projects: System R (IBM), INGRES (UCberkeley), System2000(UT-austin)
- 80's and 90's
 - Micro-DBMS family: Dbase, Foxpro, Access
 - Mainframe DBMS: DB2
 - Client-server DBMSs: ORACLE, Sybase, SQL server...

The **System R** project at IBM (1970s) led to two major developments:

1, **SQL**,

2, Commercial relational DBMS products (1980s) such as DB2 and SQL/DS from IBM and ORACLE from ORACLE

An introduction to Database Systems

- 1979年：第一个公开可用版本的Oracle数据库发布。
- 1984年：雷·奥兹（Ray Ozzie）成立Iris Associates，创造了一个受PLATO Notes启发的组合件系统。
- 1988年：由文件数据库提供支持的Lotus Agenda发布。
- 1989年：Lotus Notes发布。
- 1990年：Objectivity发布了期间对象数据库。
- 1991年：Key-value类型数据库Berkeley DB发布。
- 1991：W. H. Inmon发表了《构建数据仓库》。
- 2003年：Live Journal开放最初版本Memcached的源码。
- 2005年：达米安·卡茨（Damien Katz）开放CouchDB源码。
- 2006年：Google发表BigTable论文。
- 2007年：亚马逊发表Dynamo论文。10gen开始编制MongoDB代码。Powerset开放BigTable clone克隆版Hbase的源码。
- 2008年：Facebook开放Cassandra源码。
- 2009年：科技博客ReadWriteWeb提出一个问题：“关系型数据库是否已注定灭亡？” Redis发布。首次NoSQL会议在旧金山召开。

9/9/2025 • 2010年：Memcached项目的一些负责人与社交游戏公司Zynga开放Membase源码。

Group (Semester) project - **section 1**

Survey and describe database requirements of an organization (company, enterprise or department), briefly introduce its background,

If they have information systems?

If YES, how to use information systems? What roles that database are playing?

If NO, is there any plan to implement a database system, in which what functions that database will have ?

Describe main business processes of the surveyed object.